No	Suggested Answer (Part A)	Marks
1(a)	$\lim\limits_{x\to 2}\dfrac{(2-x)^2}{4-2x}=\lim\limits_{x\to 2}\dfrac{(2-x)^2}{2(2-x)}$ $=\lim\limits_{x\to 2}\dfrac{(2-x)}{2}$ $=\dfrac{2-2}{2}$ $=0$	K1 K1 J1
1(b)	$\lim\limits_{x\to -\infty}\dfrac{\sqrt{\dfrac{x^2+6x}{x^2}}}{\dfrac{3-2x}{-x}}=\lim\limits_{x\to -\infty}\dfrac{\sqrt{1+\dfrac{6}{x}}}{-\dfrac{3}{x}+2}$ $=\dfrac{\sqrt{1}}{2}=\dfrac{1}{2}$	K1 K1J1
1(c)	$\lim\limits_{x\to 1}\dfrac{x^2-3x+2}{\sqrt{x}-1}=\lim\limits_{x\to 1}\dfrac{(x-1)(x-2)}{\sqrt{x}-1}\times\dfrac{\sqrt{x}+1}{\sqrt{x}+1}$ $=\lim\limits_{x\to 1}\dfrac{(x-1)(x-2)(\sqrt{x}+1)}{x-1}$ $=\lim\limits_{x\to 1}(x-2)(\sqrt{x}+1)$ $=(1-2)(1+1)=-2$	K1 K1J1 9 MARKS
2a	$\dfrac{dy}{dx}=\lim\limits_{h\to 0}\dfrac{\sqrt{8(x+h)-3}-\sqrt{8x-3}}{h}$ $=\lim\limits_{h\to 0}\dfrac{\sqrt{8x+8h-3}-\sqrt{8x-3}}{h}\times\dfrac{\sqrt{8x+8h-3}+\sqrt{8x-3}}{\sqrt{8x+8h-3}+\sqrt{8x-3}}$ $=\lim\limits_{h\to 0}\dfrac{8x+8h-3-(8x-3)}{h(\sqrt{8x+8h-3}+\sqrt{8x-3})}$ $=\lim\limits_{h\to 0}\dfrac{8h}{h(\sqrt{8x+8h-3}+\sqrt{8x-3})}$	K1 K1 K1 K1

	$= \dfrac{8}{(\sqrt{8x-3}+\sqrt{8x-3})} = \dfrac{8}{2\sqrt{8x-3}} = \dfrac{4}{\sqrt{8x-3}}$	J1
2bi	$y = \ln\left(\dfrac{x-5}{x+3}\right)$ $y = \ln(x-5) - \ln(x+3)$ $\dfrac{dy}{dx} = \dfrac{1}{x-5} - \dfrac{1}{x+3}$	K1 K1J1
2bii	$y = \cos(e^{2x})$ $\dfrac{dy}{dx} = -\sin(e^{2x})\dfrac{d}{dx}\left(e^{2x}\right)$ $\dfrac{dy}{dx} = -\sin(e^{2x})\left(2e^{2x}\right)$ $\dfrac{dy}{dx} = -2e^{2x}\sin(e^{2x})$	K1 K1 J1 11 marks
3	$f(x) = 3x^4 - 4x^3 + 1$ $\Rightarrow f'(x) = 12x^3 - 12x^2$ When $f'(x) = 0, \Rightarrow 12x^3 - 12x^2 = 0$ $12x^2(x-1) = 0$ Critical points of f are $x = 0, x = 1$ By using first derivative test,	K1 J1

	$(-\infty, 0)$	$(0,1)$	$(1, \infty)$
Point in the interval	-1	0.5	2
$f'(x)$	-	-	+
	\	\	/

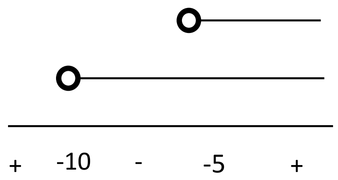
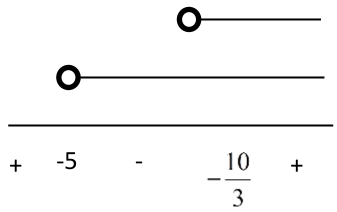
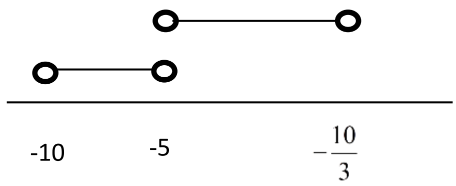
When $x = 1, f(1) = 0,$

$\Rightarrow (1,0)$ is the relative minimum point

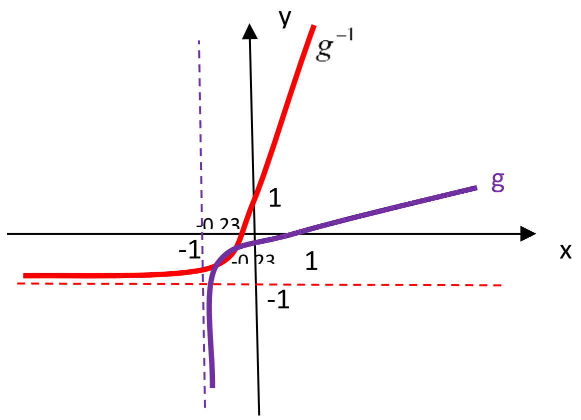
K1 K1 J1

5 marks

No	Suggested Answer (Part B)	Marks
1	$z=\frac{p+2-i}{p^2-i}$	
	$z=\frac{1-i+2-i}{(1-i)^2-i}$	K1
	$z=\frac{3-2i}{-3i}$	K1
	$z=\left(\frac{3-2i}{-3i}\right)\left(\frac{i}{i}\right)$	K1
	$z=\frac{3i-2i^2}{-3(-1)}$	
	$z=\frac{2}{3}+i$	J1
	$z=r(cos\theta+iSin\theta)$	
	$r=\sqrt{\left(\frac{2}{3}\right)^2+1^2}, r=\frac{\sqrt{13}}{3}$	K1J1
	$tan\theta=\frac{1}{\frac{2}{3}}=\frac{3}{2};\ \theta=0.983rad$	K1J1
	$z=\frac{\sqrt{13}}{3}[cos(0.983rad)+iSin(0.983rad)]$	J1
		9 marks
2a	$\left\lvert\frac{x}{5+x}\right\rvert>2$	
	$\frac{x}{x+5}>2$ or $\frac{x}{x+5}<-2$	B1
	$\frac{x}{x+5}-2>0$ or $\frac{x}{x+5}+2<0$	
	$\frac{x-2(x+5)}{x+5}>0$ or $\frac{x+2(x+5)}{x+5}<0$	K1
	$\frac{-x-10}{x+5}>0$ or $\frac{3x+10}{x+5}<0$	
	$\frac{x+10}{x+5}<0$	K1

	  $\Rightarrow(-10,-5)$ $\Rightarrow\left(-5,-\frac{10}{3}\right)$  $\left\{x:-10<x<-5\ \ or\ \ -5<x<-\frac{10}{3}\right\}$	K1 J1 K1 J1
2b	$\frac{4}{2+e^{x}}=\frac{6}{3+e^{-x}}$ $4\left(3+\frac{1}{e^{x}}\right)=6\left(2+e^{x}\right)$ $4\left(\frac{3e^{x}+1}{e^{x}}\right)=12+6e^{x}$ $12e^{x}+4=12e^{x}+6e^{2x}$ $e^{2x}=\frac{2}{3}$ $2x=\ln\left(\frac{2}{3}\right)$ $x=\frac{1}{2}\ln\left(\frac{2}{3}\right)$	K1 K1 K1 K1 J1 12 marks

3a	$f(x)=\ln(2x+3) \quad g(x)=e^{-x}+3$ $gf(x)=g(\ln(2x+3))$ $=e^{-\ln(2x+3)}+3$ $=\dfrac{1}{2x+3}+3$ $=\dfrac{10+6x}{2x+3}$ $gf\left[(gf)^{-1}(x)\right]=x$ $\dfrac{10+6(gf)^{-1}}{2(gf)^{-1}+3}=x$ $10+6(gf)^{-1}=2x(gf)^{-1}+3x$ $10-3x=(gf)^{-1}(2x-6)$ $\therefore (gf)^{-1}(x)=\dfrac{10-3x}{2x-6}$ $\therefore (gf)^{-1}(2)=\dfrac{10-3(2)}{2(2)-6}=-2$	K1 K1 K1 J1 K1J1
3b	$gg^{-1}(x)=x$ $e^{-g^{-1}(x)}+3=x$ $e^{-g^{-1}(x)}=x-3$ $-g^{-1}(x)=\ln(x-3)$ $g^{-1}(x)=-\ln(x-3)$ $g^{-1}f(x)=0$ $g^{-1}[\ln(2x+3)]=0$ $-\ln[\ln(2x+3)-3]=0$ $\ln(2x+3)-3=e^{0}$ $\ln(2x+3)=4$ $2x+3=e^{4}$ $x=\dfrac{e^{4}-3}{2}$	K1 J1 K1 K1 J1 11 marks

4	$g(x)=\frac{1}{3}\ln\left(\frac{x+1}{2}\right)$ $gg^{-1}(x)=x$ $\frac{1}{3}\ln\left(\frac{g^{-1}(x)+1}{2}\right)=x$ $\ln\left(\frac{g^{-1}(x)+1}{2}\right)=3x$ $\frac{g^{-1}(x)+1}{2}=e^{3x}$ $g^{-1}(x)=2e^{3x}-1$  $D_{g^{-1}}=\Re$ $R_{g^{-1}}=(-1,\infty)$	 K1 K1 J1 R1 R1 B1 B1 7 marks
5a	$\lim_{x\to -2} f(x)$ exists. $\lim_{x\to -2^+}(4+p)x^2 = \lim_{x\to -2^-}\frac{4}{x^2}+p$ $(4+p)(-2)^2=\frac{4}{(-2)^2}+p$ $16+4p=1+p$ $p=-5$ *f(x)* is continuous at *x*=*-2*.	 K1 K1 J1

	$\lim_{x\to-2} f(x) = f(-2)$ $\lim_{x\to-2} (4-5)x^2 = (-1)(-2)^2 = -4$ $f(-2) = -4$	K1 J1
5b	If $g(x)$ is continuous at $x = \ln 7$, then $\lim_{x\to ln7} g(x) = g(ln7)$ $\lim_{x\to ln7} g(x) = \lim_{x\to ln7} \frac{e^{2x}-49}{e^x-7}$ $= \lim_{x\to ln7} \frac{(e^x)^2-7^2}{e^x-7}$ $= \lim_{x\to ln7} \frac{(e^x-7)(e^x+7)}{e^x-7}$ $= \lim_{x\to ln7}(e^x+7)$ $= (e^{ln7}+7)$ $= 14$ $g(ln7) = 12$ $\lim_{x\to ln7} g(x) \neq g(ln7)$ Therefore $g(x)$ is not continuous at $x = \ln 7$	K1 K1 K1 J1 K1 J1 11 marks

6a	$2y\dfrac{dy}{dx}+\left(x\dfrac{dy}{dx}+y\right)-3=0$	K1K1
	$(2y+x)\dfrac{dy}{dx}=3-y$	K1
	$\dfrac{dy}{dx}=\dfrac{3-y}{2y+x}$	J1
	$\dfrac{d^2y}{dx^2}=\dfrac{(2y+x)\left(-\dfrac{dy}{dx}\right)-(3-y)\left(2\dfrac{dy}{dx}+1\right)}{(2y+x)^2}$	K1K1
	$=\dfrac{-(2y+x)\left(\dfrac{3-y}{2y+x}\right)-(3-y)\left(2\left(\dfrac{3-y}{2y+x}\right)+1\right)}{(2y+x)^2}$	
	$=\dfrac{-(2y+x)\left(\dfrac{3-y}{2y+x}\right)-(3-y)\left(\dfrac{6+x}{2y+x}\right)}{(2y+x)^2}$	K1
	$=\dfrac{-(2y+x)(3-y)-(3-y)(6+x)}{(2y+x)^3}$	
	$=\dfrac{(3-y)[-(2y+x)-(6+x)]}{(2y+x)^3}$	K1
	$=\dfrac{(3-y)(-2y-2x-6)}{(2y+x)^3}=\dfrac{(y-3)(2y+2x+6)}{(2y+x)^3}$	J1
6b	$x=3e^{2t},\quad y=3^{3t}-1$	
	$\dfrac{dx}{dt}=6e^{2t},\quad \dfrac{dy}{dt}=3^{3t}3\ln 3$	K1K1
	$\dfrac{dy}{dx}=3^{3t}3\ln 3\times\dfrac{1}{6e^{2t}}$	
	$\dfrac{dy}{dx}=\dfrac{3^{3t}\ln 3}{2e^{2t}}$	J1

	$\frac{d}{dt}\left(\frac{dy}{dx}\right)=\frac{\left(2e^{2t}\right)\left(3^{3t}3(\ln 3)^2\right)-\left(3^{3t}\ln 3\right)\left(4e^{2t}\right)}{\left(2e^{2t}\right)^2}$	K1
	$\frac{d}{dt}\left(\frac{dy}{dx}\right)=\frac{\left(2e^{2t}3^{3t}\ln 3\right)\left(3\ln 3-2\right)}{\left(2e^{2t}\right)^2}$	K1
	$\frac{d}{dt}\left(\frac{dy}{dx}\right)=\frac{3^{3t}\ln 3\left(3\ln 3-2\right)}{2e^{2t}}$	K1
	$\frac{d^2y}{dx^2}=\frac{3^{3t}\ln 3\left(3\ln 3-2\right)}{2e^{2t}}\times\frac{1}{6e^{2t}}$	J1
	$\frac{d^2y}{dx^2}=\frac{3^{3t}\ln 3\left(3\ln 3-2\right)}{12e^{4t}}$	16 marks
7.	$V=\pi r^2h$	
	$\pi r^2h=5\Rightarrow h=\frac{5}{\pi r^2}$	K1
	$A=\pi r^2+2\pi rh$	
	$A=\pi r^2+2\pi r\left(\frac{5}{\pi r^2}\right)$	K1
	$A=\pi r^2+\frac{10}{r}$	J1
	$\frac{dA}{dr}=0$	
	$2\pi r-\frac{10}{r^2}=0$	K1
	$2\pi r^3-10=0$	K1
	$\pi r^3=5$	
	$r^3=\frac{5}{\pi}$	
	$r=1.17m$	J1
	$\frac{dA}{dr}=2\pi r-\frac{10}{r^2}$	

	$\dfrac{d^2A}{dr^2} = 2\pi + \dfrac{20}{r^3}$	K1
	$= 2\pi + \dfrac{20}{\left(\dfrac{5}{\pi}\right)}$	K1
	$= 6\pi > 0$ (minimum)	
	$\therefore r_{\min} = 1.17m$	J1
		9 marks